Question 1

(a) Write 540 as a product of its prime factors.

. . . . [3]

(b) Find the lowest common multiple (LCM) of 540 and 210.

. . . . [3]

Question 2

(a) Shade the region

$A \cap B'$
MATH$_{E}ND$

ontheVenndiagram.

. . . . [3]

(b) The total number of elements in the universal set is

n(E) = 40 [MATH$_{E}ND$]
Giventhat

n(A) = 18 [MATH$_E$$ND$]
,

n(B) = 22 [MATH$_E$$ND$]
and

n(A $\cup B)' = 5[MATH_END]$
$, find$

n(A $\cap B)[MATH_END]$
.

. . . . [3]

Question 3

(a) Calculate

$\sqrt[3]{343} \times 2^{-3} \overline{5^0 [MATH_END] and write your answer as a fraction.}$

. . . . [3]

(b) Simplify

$\left(\frac{64x^6}{y^{-3}}\right)^{-\frac{2}{3}}$ $[MATH_END]$
.

. . . . [3]

Question 4

(a) Write the recurring decimal

0.2
MATH$_{E}ND$

as a fraction in its lowest terms. You must show all your working.

.... [3]

(b) Nitin scores 64 marks out of 80 in a chemistry test. In a physics test, he scores 78% . Find the test in which Nitin achieved the higher percentage and state the difference between his two percentage scores.

.... [3]

Question 5

(c) Write these quantities in order of size, starting with the smallest.

$\frac{22}{7}$ $\quad$ π $\quad$ 3.141 $\quad$ $\sqrt{9.86}$[MATH_END]

. . . . [3]

(d) Write these numbers in order of size, starting with the smallest.

2^{60} $\quad$ 3^{48} $\quad$ 5^{36} $\quad$ 6^{24}
MATH_END

. . . . [3]